

Mid Sem 20

End Sem 40

Internal 40 → 10 Attendance

HW, 2-5 hrs work studies

Books:

Principles of Software Eng. K

Edmund Clarke: Model checking.

2 Lec, 1 Tutorial

Prerequisites

How a program runs

- CS
 - ethics, safety
 - public trust
-

Energy grids

↳ power plants

↳ by PMUs, managed by software

If such software fails,

→ astx

1. Responsible computing
2. Integrity in software
3. Reliable Tech.

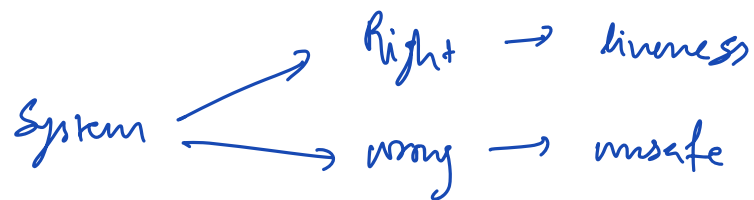
Does it work → Can we trust it?

trust efficient?

AE in Bank to approve loan $\rightarrow$ how much can we trust it?

- does it compile?
- does it pass Test Cases?

80% time is spent on testing



$\rightarrow$

Some properties are defined (mathematically),

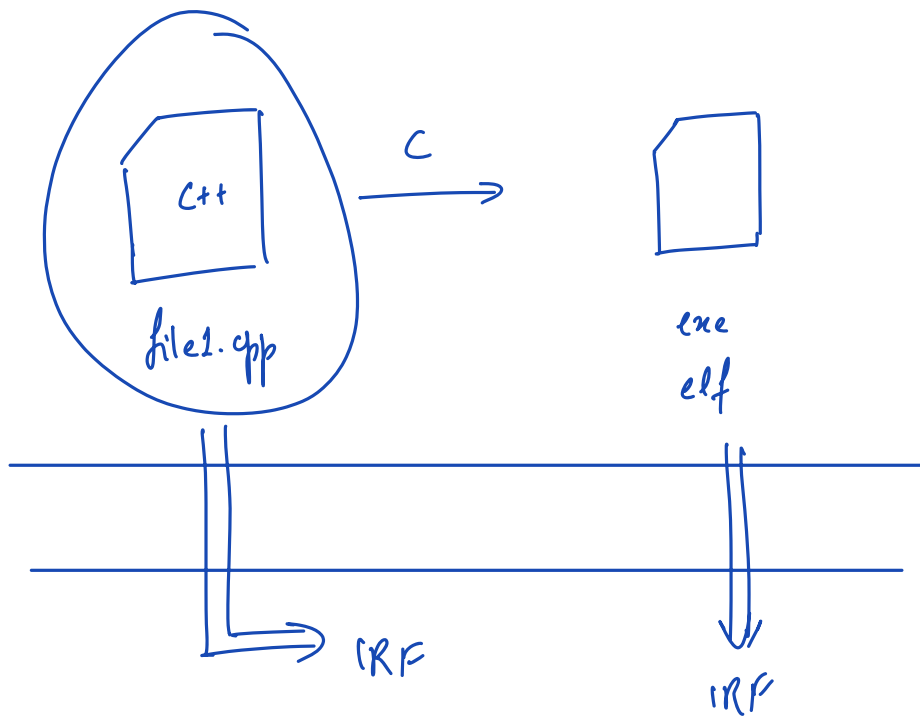
and correctness is judged if the software is correct
for these properties.

Notent-11 model / line ... & / ...

interface / module / program or / p release

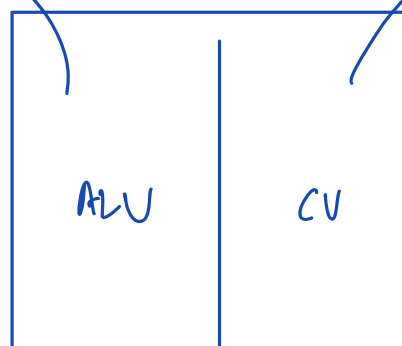
Module: Formal Verification

verification on
source code : source code
analysis



logic

control



CU → control
unit



CPV